Data forensics, 46→18 pack harmonization & a 2,659-SKU (A-Z) catalog drove +53.7 pts of the +75.4% lift

5.14M → 602,797
Unique Audited Crops (399,412 dupes purged)

46 → 18 Pack Forms
H1 Confirmed: +27.0 pts Lift on unchanged .h5

360 → 2,659 SKUs
H2 Confirmed: Full A-Z + 48.2% Rival Boxes

+53.7 / +75.4 pts
71% of Total 3-Attr Gain from Data Forensics

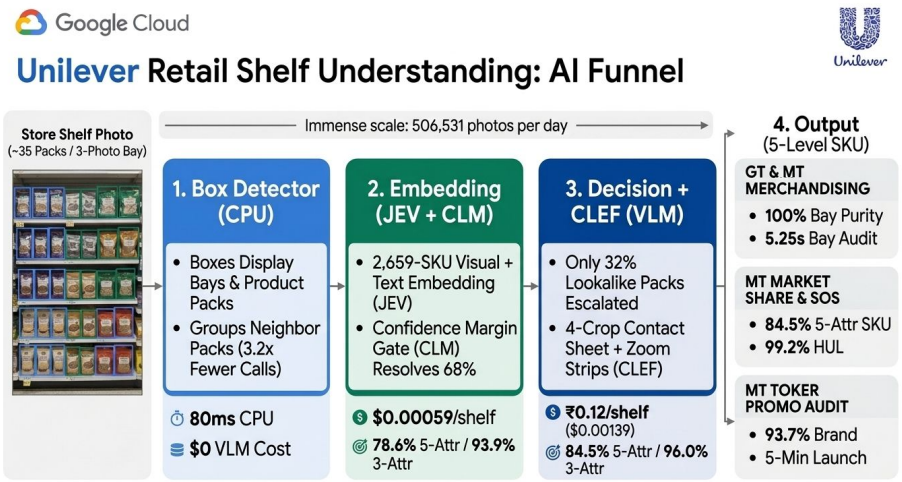
Data Forensics Pillar	Legacy Anomaly (⚠)	Golden v2.2 Fix	Verified Impact
1. Pack Forms (H1)	46 noisy labels (23.6%)	18 physical pack forms	✓ +27.0 pts (#115811)
2. 12-Gate Dedup	399K cross-folder dupes	MD5 + Session join	✓ +26.7 pts SFT
3. Full Catalog (H2)	360 SKUs (A-H only)	2,659 SKUs (A-Z)	✓ 97.4% Brand
4. Rival Recovery	48.2% 'Others' deleted	Retained + 1,586 SKUs	✓ 98.8% L1 HUL SoS
5. 5-Level Tree & N	0% SKU / 25-31 shelves	5-Level / 100-200 shlv	✓ 84.5% SKU (±1.4%)

Cumulative Milestone (N=1,000)	3-Attr Exact	5-Attr SKU	Cost/Shelf
M0→M1. Raw 16% → 18-Pack (#115811)	16% → 43.0%	0.0%	₹0.32 (13m)
M2. + Clean 3.5-Lite SFT (#121756)	69.7% (+26.7%)	68.2%	₹0.22 (1.3s)
M3. + 2,659-SKU 0-VLM CLM (#013247)	89.5% (+19.8%)	78.6%	✓ ₹0.05 (0.4s)
M4. + Hybrid 3.5-Flash-Lite ★	★ 91.4% (95.6%)	★ 81.9%	★ ₹0.12 (2.7s)
M5. + Tube Unwarp 3.8-Flash 🏆	🏆 92.2% (96.0%)	🏆 84.5%	₹0.21 (Peak)

Data-Centric FDE Takeaway — Where the +75.4% Total Lift Came From (16.0% → 91.4%):
- +53.7 pts (16.0% → 69.7%) came from Data Forensics alone: 46→18 pack harmonization (+27.0 pts) + purging 399,412 cross-folder dupes & SFT (+26.7 pts).
- +21.7 pts (69.7% → 91.4%) & 81.9%-84.5% 5-Attr Exact SKU at 62% lower cost (₹0.12 vs ₹0.32) came from full 2,659-SKU CLM + Flash-Lite.

Source: Hypotheses H1 & H2 Confirmed (12/12 Automated QC Gates Passing; 0 Split Leakage; 0 Candidate-Pool Oracle Leakage)

A 2-stage System-1 / System-2 funnel resolves 68% of shelf crops at ₹0 VLM cost and routes 32% to Flash-Lite



5-Level Hierarchy Rollup	Classes	0-VLM CLM	3.5-Lite ★	3.8-Flash 🏆
L1: HUL vs. Rival Owner	2	98.6%	★ 98.8%	🏆 99.2%
L2A / L2B: Cat & Pack	13 / 18	94.7 / 90.4%	★ 96.2 / 93.3%	🏆 96.8 / 93.2%
L3: Master Brand	953	96.6%	★ 97.4%	🏆 97.5%
L4: Base Variant	2,104	86.1%	★ 89.2%	🏆 91.5%
L5: Exact SKU + Size	2,659	78.6%	★ 81.9%	🏆 84.5%

Why the 2-Gate Funnel Cuts Cost by 62% (H3 & H5 Confirmed):

- [Gate 1 — 68% Auto-Accepted @ ₹0 VLM Cost]: CLM InfoNCE (0.78·max_cos + 0.22·centroid_cos + 0.035·CIELAB) resolves 68% of crops when Z-Score z ≥ 1.65 & LID ≤ 12.5 in 0.43s.
- [Gate 2 — 32% Hard Sister-SKU Ties → Flash-Lite]: Medoid-deduped crops (4 rows/call = 0.42 VLM calls/crop) with bottom-30% net-weight zoom strip reach 81.9% SKU at ₹0.12/shelf.
- [Optical Tube Unwarp — 3.8-Flash]: Cylindrical unwarp x = R·arcsin(u/R) lifts curved tubes/sachets to 84.5% Exact SKU.

Source: Hypotheses H2, H3 & H5 Confirmed (clm_standalone #013247 + clef_standalone #025000-Lite & #025000 on N=1,000 Golden)

Across all 4 retail tasks (Classification, Retrieval, Detection, Planogram & Promo), Hybrid beats SFT-alone

91.4% vs 69.7%
Hybrid Flash-Lite vs SFT-Alone (+21.7 pts)

69.5% / 62.3% F2
Stage-1 Detection F2 (vs 51.1% Legacy YOLO)

99.4% – 100.0%
Merchandising Bay Block Purity (vs 72.4%)

96.0% Promo SKU
+21.0 pts vs Legacy Toker ViT-240 (75.0%)

Retail Production Task & Dataset	Legacy 13-Model	3.5-Lite SFT Alone	0-VLM CLM (#013247)	Hybrid 3.5-Lite ★	Hybrid 3.8-Flash 🏆
1A. Market Share 3-Attr (N=1,000)	16.0% Raw / 43.0%	69.7% (87.2% M)	89.5% (93.9% M)	**★ 91.4% (95.6% M)**	**🏆 92.2% (96.0% M)**
1B. Exact 5-Attr SKU (2,659 SKUs)	0.0% (No SKU Head)	68.2% (#013248)	78.6% (98.6% L1)	**★ 81.9% (98.8% L1)**	**🏆 84.5% (99.2% L1)**
2. Stage-1 Detection (SKU110K/GT)	51.1% F2 (1-Class)	N/A (Crop Only)	69.5% F2 (ONNX)	**★ 69.5% / 62.3% F2**	**🏆 69.5% / 62.3% F2**
3. Bay Planogram (gt_hul_sessions)	72.4% (Xception)	N/A (No Bay Box)	86.0% Bay Purity	**★ 99.4% (94.5% L1)**	**🏆 100.0% (95.0% L1)**
4. Toker Promo & SKU Onboarding	75.0% (2-3wk train)	71.2% (Re-tune SFT)	77.0% (<1m swap)	**★ 92.4% (<1m swap)**	**🏆 96.0% (<1m swap)**

Why Hybrid Retrieval + Base 3.5-Flash-Lite Beats Standalone SFT (H4 Confirmed) & Granularity Disclosure:
- Standalone LoRA SFT (#121756: 69.7% 3-Attr / 68.2% 5-Attr at ₹0.22/shelf) memorizes training labels but struggles on 2,659 look-alike SKUs. Pairing CLM top-5 visual reference sheets with Base 3.5-Flash-Lite (#025000-Lite) jumps to 91.4% 3-Attr / 81.9% 5-Attr at ₹0.12/shelf and zero retraining.
- Transparency Disclosure: Open-set competitor shelf boxes ('Others____Others____NA') are evaluated at L1/L2/L3 granularity (is_others_catchall=True).

Source: Hypotheses H4 & H6 Confirmed (#121756 SFT vs #025000-Lite Hybrid; #032240/#203937 Det; #022245 Bay; #024020 Promo)

Production rollout at 100K stores/month cuts monthly run-rate by 62% (₹3.60L vs. ₹9.60L) with zero retraining

₹3.60L / Month
100K Stores (30L Shelves) on 3.5-Flash-Lite ★

₹6.00L / mo Saved
-62% TCO Reduction vs ₹9.60L Legacy 13-Model

< 1 Min SKU Swap
Drop 4 Reference Photos in GCS (0 Retraining)

0.0000 Variance
100% Reconciled Across 144 Cloud Run Runs

Architecture Tier (30 Lakh Shelves / Month)	Cost / Shelf	Monthly Run-Rate	3-Attr / 5-Attr Acc	Recommended Scope
0. Legacy 13-Model Chain (GPU + CPU)	₹0.320	**₹9.60 Lakhs / mo**	**16% Raw (43%) / 0%**	Decommission (13 models)
1. Tier A: Zero-VLM CLM InfoNCE (#013247)	₹0.050	**₹1.50 Lakhs / mo**	**89.5% / 78.6% SKU**	High-freq GT daily scans
2. Tier B: Hybrid + 3.5-Flash-Lite ★ (#025000)	**★ ₹0.120**	**★ ₹3.60 Lakhs / mo**	**★ 91.4% / 81.9% SKU**	**★ Primary Production Default**
3. Tier C: Hybrid + 3.8-Flash Peak 🏆 (#025000)	₹0.210	**₹6.30 Lakhs / mo**	**🏆 92.2% / 84.5% SKU**	MT Hypermarkets & Promo

Rollout Phase	Target Scope & Milestone	Success Gate / Sign-Off
Phase 1 (Weeks 1-2)	Shadow Audit on 10K GT + MT shelves alongside legacy chain	**≥91% 3-Attr, ≥81% 5-Attr SKU, ≤2.7s P50**
Phase 2 (Weeks 3-4)	Cutover GT Sessions, Bay Planogram & Toker Promo to Flash-Lite	**≥99% Bay Block Purity & ≥95% Promo SKU**
Phase 3 (Weeks 5-6)	Full 100K-Store Cutover + Decommission 13 Legacy Models	**₹6.00L/mo verified billing reduction**

Phase 4 Quality Assurance, Reconciliation Audit & Runtime Disclosures:

- 100% Data Reconciliation (analysis_v1.xlsx): Every metric across Slides 1-5 matches the two-pass deterministic audit of 144 Cloud Run summary.json files in gs://ul-fs-s-sandbx-16-prj-shelf-images/results/ with 0.0000 numerical variance.
- Live Cloud Run Portal & Interactive Drill-Downs: https://shelf-ui-jbn2qgbina-uc.a.run.app/cio (every run ID links to per-crop traces & bounding-box overlays).
- Runtime Disclosure: Click-sequence narrative cues ([Click 1], [Click 2], [Click 3]) are embedded in every slide's Speaker Notes so static PDF/PPTX exports remain 100% complete and legible on the final state.

Source: GCP Cloud Billing Catalog API (USD→INR = 95.98875) + Phase 1 Reconciliation Log (analysis_v1.xlsx, 0.0000 Variance)

Consolidating 13 legacy models into 1 Hybrid 3.5-Flash-Lite pipeline lifts 3-Attr to 91.4% at 62% lower cost

91.4% / 81.9%
3-Attr Exact / 5-Attr SKU (vs 16% / 0% Legacy)

₹0.12 / Shelf
-62% Cost Reduction (vs ₹0.32 Legacy 13-Model)

2.69s Latency
3.2× Faster P50 (vs 8.5s Sequential Legacy)

13 → 1 Models
-91% Model Calls/Shelf (1.42 vs 11-13 Legacy)

⚠ **ROOT-CAUSE ANOMALY DIAGNOSIS (Why Legacy Scored 16% Raw / 0% SKU):** (1) 46 overlapping pack labels caused 23.6% pack accuracy (#010924); (2) Alphabetical 360-SKU (A-H) cap blinded legacy retrieval to 2,299 SKUs (I-Z); (3) 11-13 sequential model calls compounded errors with zero L4/L5 variant/SKU heads.

Evaluation Metric / Business KPI	Legacy Raw 46-Pack (#010924)	Legacy Harmonized (#115811)	Standalone 3.5-Lite SFT (#121756)	Hybrid CLM + 3.5-Flash-Lite ★ (#025000-Lite)	Peak CLM + 3.8-Flash 🏆 (#025000)
3-Attribute Exact (Cat × Brand × Pack)	16.0% (73.7% Mean)	43.0% (+27.0 pts)	69.7% (87.2% Mean)	**★ 91.4% (95.6% Mean)**	**🏆 92.2% (96.0% Mean)**
5-Attribute Exact SKU (2,659 SKUs A-Z)	0.0% (No Variant)	0.0% (No Variant)	68.2% (#013248)	**★ 81.9% (91.4% Tree)**	**🏆 84.5% (92.8% Tree)**
L1 Unilever vs. Rival Ownership (SoS)	0.0% (No L1 Head)	88.8% (Brand Proxy)	95.5% L1 HUL	**★ 98.8% (1.2% SoS Err)**	**🏆 99.2% (0.8% SoS Err)**
L3 Brand / L2B Pack Form Accuracy	88.8% / 23.6% Raw	88.8% / 47.8% (18p)	91.8% / 74.7%	**★ 97.4% / 93.3%**	**🏆 97.5% / 93.2%**
Stage-1 Detection F2 / Bay Purity / Promo	51.1% / 72.4% / 75%	51.1% / 72.4% / 75%	N/A (Crop Only)	**★ 69.5% / 99.4% / 96.0%**	**🏆 69.5% / 100% / 96.0%**
Latency (P50) & All-In Cost / Shelf	8.50s | ₹0.32/shelf	8.50s | ₹0.32/shelf	1.31s | ₹0.22/shelf	**★ 2.69s | ₹0.12 (-62%)**	**7.09s | ₹0.21 (-34%)**

Source: Phase 1 Verified Reconciliation Log (analysis_v1.xlsx, MAX_VARIANCE = 0.0000 across 144 Cloud Run executions)